

# Pricing plans for Azure Quantum providers

In Azure Quantum, hardware and software providers define and control the pricing of their offerings. The information in this article is subject to change by providers and some delays in reflecting latest pricing information may exist. Be sure to verify the latest pricing information from the Azure Quantum workspace that you're using.

## ▸ IonQ

[IonQ](#) charges based on a token pricing model which billing unit is the *Azure Quantum Token* (AQT). This model is specific to Azure Quantum. The number of Azure Quantum Tokens is calculated by the following formula:

$$AQT = m + 0.000220 \cdot (N_{1q} \cdot C) + 0.000975 \cdot (N_{2q} \cdot C)$$

where:

- $AQT$  is the number of Azure Quantum Tokens consumed by the program
- $m$  is the minimum price per program execution, which is USD97.50 if error mitigation is on and USD12.4166 if error mitigation is off
- The units  $N_{1q} \cdot C$  and  $N_{2q} \cdot C$  are called *gate-shots*, where  $N_{1q}$  and  $N_{2q}$  are the number of one- and two-qubit gates submitted, respectively, and  $C$  is the number of execution shots

Multi-controlled two-qubit gates are billed as  $6 * (N - 2)$  two-qubit gates, where  $N$  is the number of qubits involved in the gate. For example, a NOT gate with three controls would be billed as  $(6 * (4 - 2))$  or 12 two-qubit gates. One-qubit gates are billed as 0.225 of a two-qubit gate (rounded down). To learn more about IonQ, visit [IonQ provider page](#).

IonQ offers a **pay-as-you-go** plan and a **monthly subscription** plan with access to the quantum simulator, the Aria 1 25-qubit quantum computers, and IonQ Forte 1 and Forte Enterprise 1 36-qubit quantum computers.

### Pay As You Go

The Pay-as-you-go plan consists of *a la carte* access to the IonQ Aria 1 25-qubit, IonQ Forte 1 and Forte Enterprise 1 36-qubit quantum computers, and the IonQ simulator. The use of the quantum computers is charged based on the number of AQTs + Azure infrastructure costs.

Expand table

| Includes access to      | Pricing                                                                                                                                                                                                                                                                                                                           |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IonQ simulator          | Free of charge                                                                                                                                                                                                                                                                                                                    |
| IonQ Aria 1             | <ul style="list-style-type: none"><li>• USD0.000220 / 1-qubit-gate shot</li><li>• USD0.000975 / 2-qubit-gate shot</li><li>• Minimum price per program execution:<ul style="list-style-type: none"><li>◦ USD97.50 - default setting, error mitigation is on</li><li>◦ USD12.4166 if error mitigation is off</li></ul></li></ul>    |
| IonQ Forte 1            | <ul style="list-style-type: none"><li>• USD0.0001645 / 1-qubit-gate shot</li><li>• USD0.001121 / 2-qubit-gate shot</li><li>• Minimum price per program execution:<ul style="list-style-type: none"><li>◦ USD168.195 - default setting, error mitigation is on</li><li>◦ USD25.7899 if error mitigation is off</li></ul></li></ul> |
| IonQ Forte Enterprise 1 | <ul style="list-style-type: none"><li>• USD0.0001645 / 1-qubit-gate shot</li><li>• USD0.001121 / 2-qubit-gate shot</li><li>• Minimum price per program execution:<ul style="list-style-type: none"><li>◦ USD168.195 - default setting, error mitigation is on</li><li>◦ USD25.7899 if error mitigation is off</li></ul></li></ul> |

For more information about Azure infrastructure costs, see [Azure Blob Storage pricing](#) .

ⓘ **Note**

If you are an existing research grant or a Pay-as-you-go customer, you may see another billing unit in addition to AQT, called *Quantum Gate-Shots (QGS)*. The QGS units are equivalent to AQT units. The number of QGS is calculated by the following formula:

$$QGS = N \cdot C$$

where:

- *QGS* is the number of quantum gate-shots
- *N* is the number of one- or two-qubit gates submitted
- *C* is the number of execution shots

# Pasqal

**Pasqal** charges for job execution time on its quantum processor - the 100-qubit Orion Beta generation called Fresnel - and its state-of-the-art emulators - EMU-MPS and EMU-SV.

Pasqal offers one billing plan: **Pay As You Go**.

Pay As You Go

In the Pay-as-you-go plan the usage is charged based on the job execution time only.

Expand table

| Pricing                                                                                                                                                               | Includes access to                                                                                                             |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• EUR 3,000 per QPU-hour + Azure infrastructure costs</li><li>• EUR 15 per emulator-hour + Azure infrastructure costs</li></ul> | <ul style="list-style-type: none"><li>• Pasqal FRESNEL, FRESNEL_CAN1 QPUs</li><li>• Pasqal EMU-MPS, EMU-SV emulators</li></ul> |

For more information about Azure infrastructure costs, see [Azure Blob Storage pricing](#) .

## Quantinuum

**Quantinuum** uses a system that charges each job depending on the number of operations in the job, and the number of shots you run. The usage units defined by Quantinuum are *Hardware Quantum Credits (HQCs)* for jobs submitted to quantum computers and emulator HQCs (eHQCs) for jobs submitted to emulators.

HQCs and eHQCs are used to calculate the cost of running a job, and they are calculated based on the following formula:

$$HQC = 5 + C(N_{1q} + 10N_{2q} + 5N_m)/5000$$

where:

- $N_{1q}$  is the number of single-qubit operations in a circuit.
- $N_{2q}$  is the number of native two-qubit operations in a circuit. Native gate is equivalent to CNOT up to several single-qubit gates.
- $N_m$  is the number of state preparation and measurement (SPAM) operations in a circuit including initial implicit state preparation and any intermediate and final measurements and state resets.
- $C$  is the shot count.

To learn more about Quantinuum, visit the [Quantinuum provider page](#).

Quantinuum provides two subscription plans: **Standard** and **Premium**. Quantinuum also offers a **Pay as You Go** offering.

Subscriptions

Subscriptions are monthly subscription plans available through queued access.

Expand table

| Plans and Pricing                                                   | Includes access to                                                                                                                                                    |
|---------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Standard Plan:</b> USD125,000/Month + Azure infrastructure costs | <ul style="list-style-type: none"><li>10k HQCs for use on Quantinuum System Model H2</li><li>100k eHQCs for use on the Quantinuum System Model H2 emulators</li></ul> |
| <b>Premium Plan:</b> USD175,000/Month + Azure infrastructure costs  | <ul style="list-style-type: none"><li>17k HQCs for use on Quantinuum System Model H2</li><li>170k eHQCs for use on the Quantinuum System Model H2 emulators</li></ul> |

For more information about Azure infrastructure costs, see [Azure Blob Storage pricing](#) .

## Rigetti

[Rigetti](#) charges for job execution time on their quantum processors. There's no added charge per job, per shot, or per gate. The [Quantum Virtual Machine \(QVM\)](#) simulator is free for all users under all plans.

Rigetti offers one billing plan: **Pay As You Go**.

### Pay As You Go

The Pay-as-you-go plan consists of *a la carte* access to Rigetti QPUs. The usage is charged based on the job execution time only.

Expand table

| Pricing                                                     | Includes access to     |
|-------------------------------------------------------------|------------------------|
| USD 0.02 per 10-millisecond increment of job execution time | Rigetti Cepheus-1-108Q |

 **Note**

If you have questions or run into any issue using Azure Quantum, you can contact [AzureQuantumInfo@microsoft.com](mailto:AzureQuantumInfo@microsoft.com).

# Related content

- [Azure Quantum quotas](#)
- [Quantum computing target list](#)

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